

Submission map

1. Fully functional source code

A single Python codebase covering the EnergyPlus API wrapper, the LLM agent orchestration, and the communication bus.

CONCERN	FILE
EnergyPlus API wrapper	<code>src/eplus_runner.py</code> — in-process <code>pyenergyplus</code> runner, sensing + actuation callbacks, handle resolution
LLM agent orchestration	<code>src/agent.py</code> — prompting, JSON schema, self-correction, decision cache, fallback
Communication bus	<code>src/mcp_server.py</code> (5 MCP tools, stdio) · <code>src/mcp_bridge.py</code> (sync↔async bridge) · <code>src/shared_state.py</code> (atomic cross-process state)
Safety layer	<code>src/policy.py</code> — clamp + occupancy-aware operating envelope
Experiment driver	<code>src/run_experiment.py</code> — all three modes, one command

Run everything with:

```
python src/run_experiment.py --mode all
```

2. Building models (.idf)

FILE	WHAT IT IS
<code>models/baseline.idf</code>	Base file — untouched <code>5ZoneAirCooled</code> from EnergyPlus ExampleFiles
<code>models/simulation.idf</code>	Modified at build time: run period trimmed, timestep outputs instrumented
<code>models/agent_optimized.idf</code>	Modified during runtime evaluation — the agent's applied setpoints reconstructed into <code>Schedule:Compact</code> objects

`agent_optimized.idf` exists because actuating the API leaves no file behind — setpoints live only in memory during the run. `src/export_idf.py` reads what the agent actually applied at each of 2,016 timesteps, takes the median per (day type, hour), and writes a model that runs standalone with **no LLM, no MCP server and no Python in the loop**.

Verified: it runs on its own and saves **3.79 %** with 0.00 % comfort violations.

3. Quantitative savings dashboard

`dashboard/report.html` — self-contained, no server, no network.

Rebuild: `python dashboard/build_report.py` · **Export to PDF:** open it and press Ctrl+P → Save as PDF.

Explicit proof of kWh reduction with comfort maintained:

CONTROLLER	TOTAL KWH	REDUCTION	HVAC KWH	HVAC REDUCTION	PEAK KW	COMFORT VIOLATIONS
Baseline (stock schedules)	3059.74	—	904.23	—	20.22	0.00 %
Rule-based (no LLM)	2990.15	+2.27 %	832.82	+7.90 %	19.65	0.00 %
AI closed-loop agent	2958.16	+3.32 %	800.20	+11.50 %	19.46	0.00 %

Comfort is defined as the share of **occupied** zone-timesteps outside 20–25 °C, measured across all five conditioned zones. The unconditioned plenum is excluded.

Raw data export: `results/summary.json`, `results/*/metrics.json`, `results/*/timeseries.csv`.

4. System architecture document

`README.md § Architecture` — the four required sections are its four subheadings:

1. **Tool-calling architecture** — MCP diagram, the 5 tools, why supervisory rather than per-timestep control, the cross-process bus
2. **Prompt engineering strategy** — teaching the domain rather than dictating answers, JSON schema enforcement, history compaction, context-aware constraints, self-correction
3. **Prompt latency management** — the arithmetic, supervisory cadence, decision cache, measured figures
4. **Handling lengthy simulation logs** — filter → deduplicate → truncate, with measured before/after token counts